

A photograph of a modern, minimalist building with white walls and a flat roof. The building features a series of steps leading up to a recessed entrance. There are some small, square concrete planters and a few trees in the foreground. The sky is a pale, clear blue.

Bank Customer Churn Prediction: Machine Learning Solution

Sharmila Devi Sree KS | Internship Project

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Executive Summary

This project delivers a comprehensive machine learning solution for predicting customer churn in retail banking, enabling proactive retention strategies and improved customer lifetime value.

Business Challenge

Customer retention is critical in competitive banking; traditional reactive analysis occurs too late to prevent churn

Solution Approach

Developed predictive model using supervised machine learning on European Bank customer dataset

Key Outcome

Gradient Boosting model achieved 86.9% accuracy and 0.8682 ROC-AUC score

Business Impact

Enables early identification of at-risk customers for targeted retention campaigns

Deployment

Interactive Streamlit dashboard for real-time churn prediction and risk assessment

Problem Statement and Objectives

Customer churn directly impacts revenue, customer lifetime value, and profitability in banking institutions.

Business Challenge	Impact	Solution
Customers switching to competitors	Revenue loss and reduced CLV	Predictive identification of churn risk
Reactive post-churn analysis	Unable to prevent customer loss	Proactive early warning system
Manual analysis of large datasets	Inefficient resource allocation	Machine learning automation
Broad retention campaigns	High marketing costs with low ROI	Personalized, targeted interventions



Dataset and Features

The European Bank customer dataset comprises 10,000 records with 13 predictor variables capturing demographic, financial, and behavioral customer information.

Feature Category	Variables	Description
Demographic	Age, Gender, Geography	Customer profile information
Financial	Balance, EstimatedSalary, CreditScore	Financial position indicators
Behavioral	Tenure, IsActiveMember, HasCrCard	Engagement and product usage
Product Usage	NumOfProducts	Banking products owned
Target Variable	Exited	Churn status (1=Churned, 0=Retained)



No missing values identified



No duplicate records present

8

13 original features + 4 engineered features = 17 total predictors



Binary classification problem with moderate class imbalance

Methodology and Preprocessing

Preprocessing

Removed non-informative features (CustomerId, Surname); applied one-hot encoding to categorical variables (Geography, Gender); standardized numerical features using StandardScaler

One-hot encoding for categorical variables (Geography, Gender)

StandardScaler normalization for numerical features

Feature Engineering

Created Balance-to-Salary Ratio, HasMultipleProducts indicator, Tenure-Product Interaction, and Age-Tenure Interaction features

Stratified sampling to maintain class balance across train-test datasets

Feature engineering to capture hidden behavioral relationships

Train-Test Split

80:20 stratified split preserving original class distribution

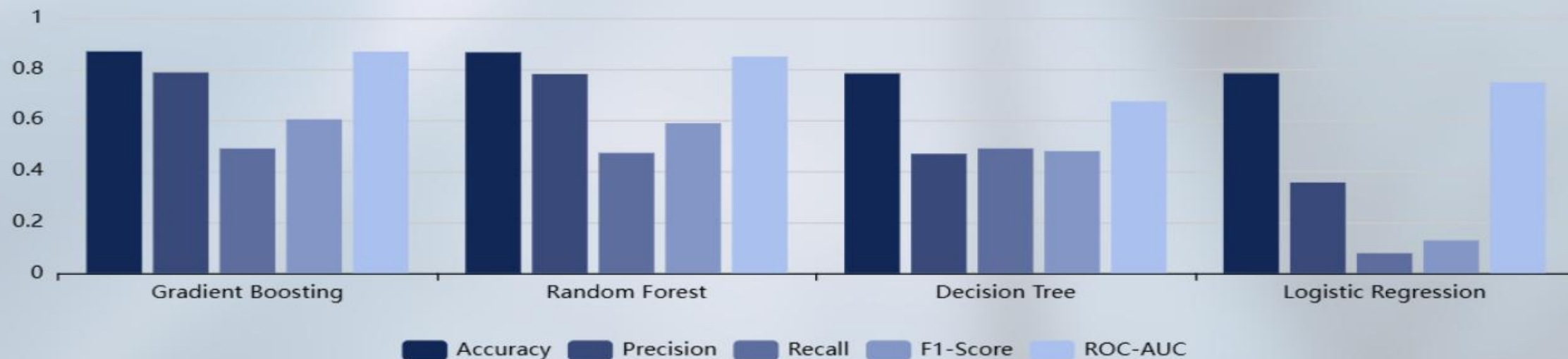
Model Training

Implemented four supervised classification algorithms for comparative analysis

Model Development and Performance

Four machine learning algorithms were developed and compared using identical evaluation metrics.

Model Performance Metrics



Gradient Boosting achieved highest ROC-AUC score (0.8682)



Maintained strong balance between Precision (0.7866) and Recall (0.4889)



Demonstrated superior generalization capability over other ensemble methods



Selected as final production model for deployment

Key Findings and Feature Importance

Explainable AI techniques identified the primary drivers of customer churn.

38.5%

Age (38.5% importance)

Older customers show higher churn tendency

28.5%

Number of Products (28.5% importance)

Multi-product customers demonstrate stronger retention

7.6%

Active Member Status (7.6% importance)

Inactive accounts indicate higher churn risk

6.2%

Account Balance (6.2% importance)

Lower balances correlate with higher churn probability

5.9%

Geography (5.9% importance)

Germany shows highest churn rates among covered regions

1

Feature Importance ranking from Gradient Boosting model

2

SHAP (SHapley Additive exPlanations) for local prediction explanations

3

Partial Dependence Plots showing feature-churn relationships

Business Insights and Recommendations

Older customers (Age > 50) with single banking products

Inactive members with low account activity

Customers with low balance-to-salary ratios

New customers with limited tenure

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Target Engagement

Offer multi-product bundles to increase switching costs and engagement

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Personalized Communication

Develop age-specific retention campaigns emphasizing loyalty benefits

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Proactive Outreach

Monitor inactive accounts and re-engagement initiatives

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Service Enhancement

Improve digital banking features and customer support for high-risk segments

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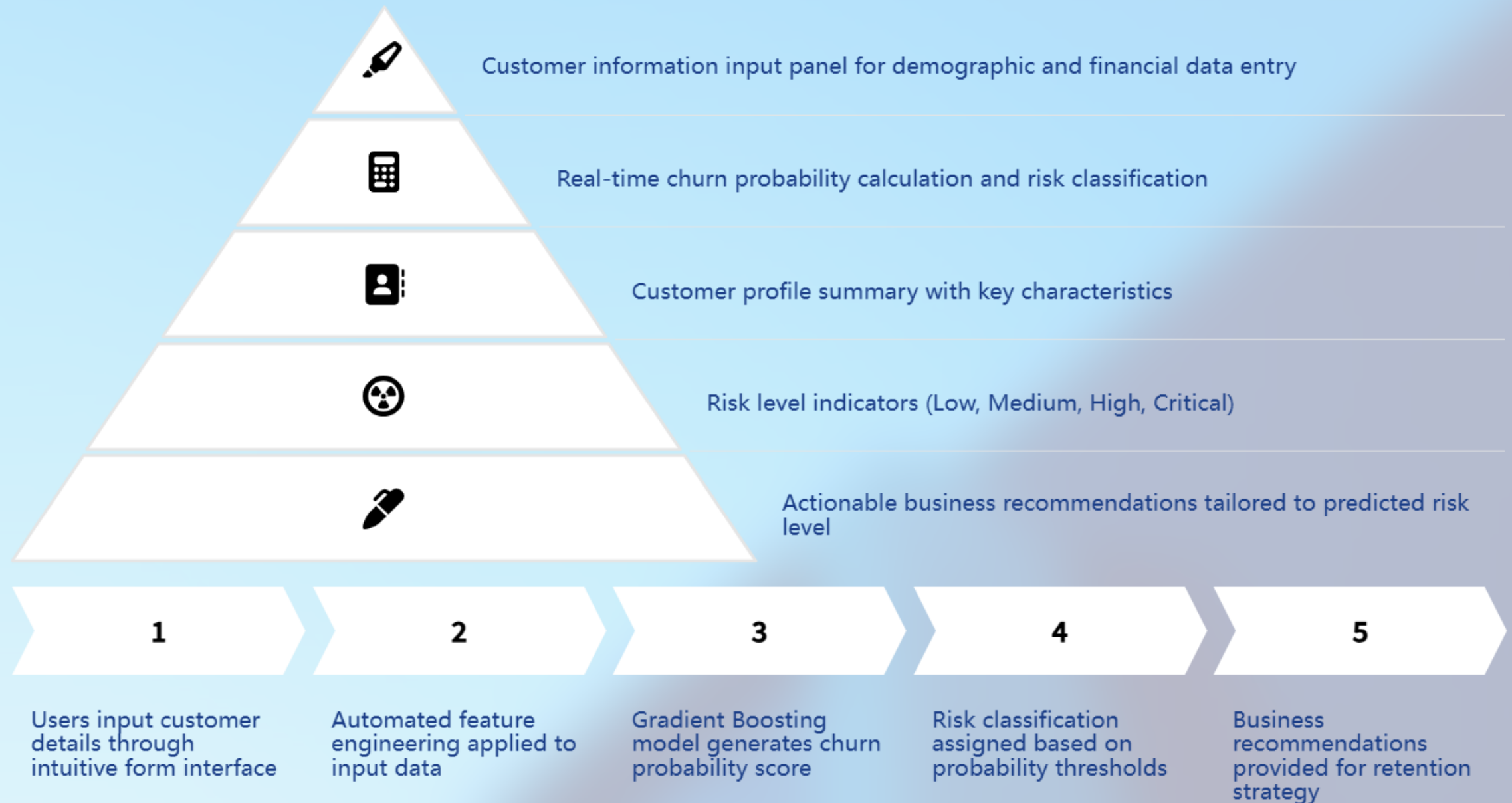
Incentive Programs

Implement rewards for product cross-selling and account activity

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Deployment and Implementation

The Gradient Boosting model was deployed as an interactive Streamlit web application enabling real-time customer churn prediction.



Business Value and ROI

Revenue Protection

Early identification reduces customer acquisition cost burden by improving retention

Enable proactive customer retention strategies before churn occurs

Resource Optimization

Target retention efforts on high-risk customers rather than broad campaigns

Support targeted marketing campaigns focused on high-risk segments

Customer Lifetime Value

Proactive interventions strengthen long-term customer relationships

Improve retention rate through early intervention mechanisms

Data-Driven Decisions

Prediction scores support objective prioritization of retention efforts

Enhance customer satisfaction through personalized engagement approaches

Operational Efficiency

Automated workflow reduces manual churn analysis time

Technical Implementation Details

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Python Libraries

Scikit-learn (model development), Pandas (data processing), NumPy (computation)

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Explainability Tools

SHAP for local/global explanations, Feature Importance ranking

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“

Deployment

Streamlit web framework for interactive dashboard

”

“

Model Format

Gradient Boosting Classifier with optimized hyperparameters

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Algorithm

Gradient Boosting Classifier (sequential decision tree ensemble)

Input Features

17 variables (13 original + 4 engineered)

Output

Binary classification (Churn probability score 0-1)

Performance Threshold

Optimized for balance between false positives and false negatives

Future Enhancement Opportunities

The solution can be further advanced through strategic improvements.



Live Database Integration

Connect to real-time banking systems for continuous prediction



Cloud Deployment

Migrate to AWS, Google Cloud, or Azure for enterprise scalability



Model Retraining

Implement automated retraining pipeline with newly available customer data



CRM Integration

Embed predictions into existing Customer Relationship Management systems



Deep Learning

Evaluate neural networks and LSTM models for sequential customer behavior



Recommendation Engine

Develop personalized retention offer recommendation system

Conclusion

This project successfully demonstrates the application of machine learning to address a critical banking challenge. The developed Gradient Boosting model achieves strong predictive performance (86.9% accuracy, 0.8682 ROC-AUC) while maintaining interpretability through explainable AI techniques. Deployment via interactive Streamlit dashboard transforms technical predictions into actionable business intelligence, enabling data-driven customer retention strategies that improve profitability and long-term customer relationships in competitive banking environments.